LESSON 2

Reading Schematics, Datasheets, and Practical Troubleshooting

NLC STEM Club UAV Project
Electrical Systems Education Series

PREREQUISITES:

- Completed Lesson 1: Circuit Fundamentals Teaching Guide
- Understanding of Ohm's Law, power calculations, and wire gauge selection
- Basic familiarity with your project components

LESSON OBJECTIVES:

By the end of this lesson, you will be able to:

1. Read and interpret electrical schematics and wiring diagrams
2. Extract critical information from component datasheets
3. Use a multimeter to measure voltage, current, and continuity
4. Understand PWM signals and how they control servos and ESCs
5. Troubleshoot common electrical problems systematically
6. Recognize and prevent grounding issues and electrical noise
7. Apply these skills to debug your UAV electrical system

1 Reading Electrical Schematics

1.1 What is a Schematic?

A schematic is a symbolic representation of an electrical circuit that shows:

- How components are electrically connected
- The flow of current and signals
- Component values and part numbers

KEY DIFFERENCE: Schematic vs Wiring Diagram

SCHEMATIC	WIRING DIAGRAM
Shows ELECTRICAL connections (how electricity flows)	Shows PHYSICAL connections (which wire goes where)
Uses standardized symbols	Shows actual connectors and wire colors
Does NOT show physical layout	Shows spatial relationships between components
Used by engineers to design and analyze circuits	Used during assembly and troubleshooting

1.2 Common Schematic Symbols

POWER SOURCES:

- **Battery:** Two parallel lines (longer line is positive)
- **DC Power Supply:** Box with V+ and GND labels

CONNECTIONS:

- **Wire connection:** Dot indicates electrical connection
- **Wires crossing:** No dot = no connection, just crossing
- **Ground symbols:** Earth ground, chassis ground, signal/digital ground

PASSIVE COMPONENTS:

- **Resistor:** Zigzag line or rectangle
- **Capacitor:** Two parallel lines (electrolytic has + polarity)
- **Inductor:** Coil symbol

ACTIVE COMPONENTS:

- **Diode:** Triangle with line (current flows in arrow direction)
- **LED:** Diode with small arrows pointing out
- **Transistor (NPN):** Three terminals: Base, Collector, Emitter

- **Voltage Regulator:** Box with IN, OUT, and GND pins

MOTORS & MECHANICAL:

- **Motor (DC):** Circle with M inside
- **Motor (Servo):** Box with mechanical coupling

SWITCHES & SENSORS:

- **Switch:** SPST (Single Pole Single Throw)
- **Button:** Normally open pushbutton

1.3 Reading a Schematic: Step-by-Step

STEP 1: Identify the Power Sources

- Look for battery symbols or "V+" labels
- Note the voltage level (e.g., 14.8V, 5V, 3.3V)
- Locate all ground symbols

STEP 2: Follow the Power Rails

- Trace from positive terminal through components to ground
- Identify parallel branches (components sharing same voltage)
- Identify series components (current flows through both)

STEP 3: Identify Signal Paths

- Find input signals (from sensors, receivers)
- Trace signal wires to processing units (Arduino, flight controller)
- Find output signals (to ESC, servos, LEDs)

STEP 4: Note Component Values

- Resistor values (Ω)
- Capacitor values (μF, pF)
- Component part numbers or ratings

EXAMPLE - Simple LED Circuit:
Current flows: 5V $\rightarrow$ through R1 (220Ω) $\rightarrow$ through LED $\rightarrow$ to GND
HOW TO READ THIS:

1. Power source: 5V
2. Current flows from 5V through resistor through LED to ground
3. Resistor limits current to safe level for LED
4. Voltage drop: 5V - 2V (LED) = 3V across resistor
5. Current: $I = V/R = 3V/220\Omega = 13.6$ mA

1.4 UAV System Schematic Example

Let's analyze a simplified version of YOUR dual-BEC system:

```
[4S LiPo Battery:  14.8V]
          ↓
     XT60 Y-Harness
      ↙          ↘
ESC              External UBEC
  5V @ 3A            5V @ 6A
   RAIL 1            RAIL 2
    ↓                 ↓
  Flight Critical    Servos
        Components
```

KEY INSIGHT

This schematic answers "WHY two BECs?"

- Total servo current could exceed ESC BEC rating
- Separating rails prevents servo noise from affecting flight controller
- If servos brown out Rail 2, Rail 1 stays stable

2 Reading Datasheets

2.1 What is a Datasheet?

A datasheet is a technical document provided by the manufacturer that contains:

- Electrical specifications (voltage, current, power)
- Mechanical dimensions and pinouts
- Performance characteristics (speed, accuracy, range)
- Operating conditions (temperature, humidity)
- Application guidelines and example circuits

DATASHEETS ARE ESSENTIAL FOR:

- Verifying compatibility with your system
- Determining proper operating conditions
- Connecting components correctly (pinouts)
- Troubleshooting when things don't work

2.2 Key Sections to Look For

SECTION 1: ABSOLUTE MAXIMUM RATINGS

WARNING

CRITICAL - exceeding these can destroy the component!
Example (from a typical voltage regulator datasheet):

- Input Voltage: 4.5V to 40V
- Output Current: 3A continuous, 5A peak
- Operating Temperature: -40°C to +125°C
- Storage Temperature: -65°C to +150°C

If you apply 50V to this regulator = instant damage!

SECTION 2: RECOMMENDED OPERATING CONDITIONS

These are SAFE ranges for normal operation

Example:

- Input Voltage: 7V to 30V (within absolute max)
- Output Current: 0 to 3A (with heatsink if > 1.5A)
- Operating Temperature: 0°C to 85°C

Design your system to stay within these ranges!

SECTION 3: ELECTRICAL CHARACTERISTICS

Detailed performance specifications

Parameter	Condition	Min	Typ	Max	Unit
Output Voltage	Vout = 5V	4.9	5.0	5.1	V
Line Regulation	Vin = 7V-30V	-	10	50	mV
Load Regulation	Iout = 0-3A	-	20	100	mV
Quiescent Current	No load	-	10	15	mA
Efficiency	Vin=12V, Iout=1A	-	85	-	%

HOW TO READ THIS:

- "Typ" = typical value (what you'll usually measure)
- "Min/Max" = guaranteed range (your unit will fall within this)
- Output voltage will be $5.0V \pm 0.1V$ (4.9V to 5.1V)

SECTION 4: PIN CONFIGURATION (PINOUT)

Shows which pin does what

WARNING

Connecting wrong pins = component destruction!

SECTION 5: APPLICATION CIRCUIT

Example schematic showing proper use. Often includes:

- Input/output capacitors (for stability)
- Protection diodes
- LED indicators
- Recommended wire gauges

SECTION 6: PACKAGE INFORMATION

Physical dimensions, hole spacing, mounting

2.3 Real-World Datasheet Example: BME280 Sensor

Let's extract key information for YOUR air quality sensor project:

FROM THE BME280 DATASHEET:

ABSOLUTE MAXIMUM RATINGS

- Supply Voltage (Vdd): -0.3V to 4.25V
- ESD Tolerance: 2000V

→ Connect to Arduino 3.3V output (NOT 5V!) to stay safe OR use a level shifter module

OPERATING CONDITIONS

- Supply Voltage: 1.71V to 3.6V
- Supply Current: 3.6 μA @ 1 Hz (typical), 714 μA @ 1 Hz (max)

→ Very low power! Won't strain your Arduino power budget

MEASUREMENT RANGES & ACCURACY

- Temperature: -40°C to +85°C (±1°C accuracy)
- Humidity: 0% to 100% RH (±3% accuracy)
- Pressure: 300 to 1100 hPa (±1 hPa accuracy)

→ Perfect for your environmental monitoring mission!
→ Can detect altitude changes of ∼8 meters

I2C INTERFACE

- I2C Address: 0x76 or 0x77 (selectable)
- Clock Speed: Up to 3.4 MHz
- Logic Levels: VddIO (0.3 × Vdd to Vdd)

→ Use Wire.begin() in Arduino code
→ Check address with I2C scanner if not responding

PINOUT (8-pin package)

- Pin 1: GND - Ground
- Pin 2: CSB - Chip Select (I2C: connect to Vdd)
- Pin 3: SDI - I2C Data (SDA)
- Pin 4: SCK - I2C Clock (SCL)
- Pin 5: SDO - Address select (GND=0x76, Vdd=0x77)
- Pin 6: Vdd - Power supply 1.71V-3.6V
- Pin 7: VddIO - Logic level reference (can = Vdd)
- Pin 8: GND - Ground

WIRING TO ARDUINO:

BME280 Breakout Board	**Arduino Nano**
VCC/Vin	3.3V (NOT 5V!)
GND	GND
SDA	A4 (I2C Data)
SCL	A5 (I2C Clock)

2.4 Practice: Extract Info from Your Component Datasheets

For EACH component in your system, find and record:

ESC DATASHEET:

1. Continuous current rating: _____ A
2. Input voltage range: _____ V to _____ V
3. BEC output voltage: _____ V
4. BEC output current: _____ A
5. PWM signal frequency: _____ Hz
6. Wiring: Red = _____, Black = _____, White/Yellow = _____

UBEC DATASHEET:

1. Input voltage range: _____ V to _____ V
2. Output voltage: _____ V (adjustable? Y/N)
3. Continuous output current: _____ A
4. Peak output current: _____ A
5. Efficiency at your voltage: _____ %

FLIGHT CONTROLLER (APM 2.8) DATASHEET:

1. Input voltage: _____ V (connect to which BEC?)
2. PWM output voltage: _____ V
3. Number of output channels: _____
4. GPS connector pinout: TX = pin _____, RX = pin _____
5. Telemetry port: _____

3 Using a Multimeter

3.1 Multimeter Basics

A multimeter measures three fundamental electrical properties:

- **VOLTAGE (V or VDC/VAC)** - Potential difference
- **CURRENT (A or mA)** - Flow of electrons
- **RESISTANCE (Ω)** - Opposition to current

PLUS useful functions:

- **CONTINUITY TEST** (beep when connected)
- **DIODE TEST** (measures forward voltage drop)

DIAL POSITIONS (typical multimeter):

- **OFF** - Power off
- **V∼ (ACV)** - AC voltage (wall outlets: 120V)
- **V- (DCV)** - DC voltage ← YOU'LL USE THIS MOST
- Ω - Resistance
- **Continuity** - Continuity test (beeper)
- **Diode** - Diode test
- **A** - Current (often separate jack!)

PROBE PLACEMENT:

- RED probe: Positive terminal, higher voltage point
- BLACK probe: Negative terminal, ground, lower voltage point

3.2 Measuring Voltage

Voltage is measured IN PARALLEL (across a component)

PROCEDURE:

1. Set dial to "V-" (DC voltage)
2. Select range: 20V for most UAV measurements
3. Touch RED probe to positive point
4. Touch BLACK probe to ground/negative point
5. Read the display

EXAMPLE 1: Checking Battery Voltage
Battery terminals: (+) and (-)
Place RED probe on (+), BLACK probe on (-)
Expected reading: 16.8V (fully charged 4S)
Safe minimum: 13.2V (3.3V per cell × 4)

EXAMPLE 2: Checking BEC Output
BEC output connector:
Red wire (+5V) - RED probe
Black wire (GND) - BLACK probe
Expected reading: $5.0V \pm 0.1V$
If reading $< 4.8V \rightarrow$ Problem! BEC may be overloaded or failing

EXAMPLE 3: Voltage Drop Across Wire
Sometimes you want to check if a wire has too much resistance:
Battery (+) at Point A: 14.8V
Component (+) at Point B: 14.5V
Voltage drop: 0.3V ← This is lost as heat in the wire!
→ If drop > 0.5V, use thicker wire gauge

WARNING

SAFETY NOTE: Never measure voltage with power off—it will read 0!

3.3 Measuring Current

WARNING

MOST DANGEROUS MEASUREMENT - CAN BLOW FUSE OR DAMAGE MULTIMETER!

Current is measured IN SERIES (breaking the circuit)

PROCEDURE:

1. Turn OFF power first!
2. Move RED probe to "A" or "10A" jack (check meter's max current!)
3. Set dial to "A" (current)
4. BREAK the circuit at one point
5. Connect multimeter IN the current path (multimeter completes the circuit)
6. Turn power ON
7. Current flows THROUGH the meter - read display
8. Turn power OFF before disconnecting!

WARNING

NEVER CONNECT MULTIMETER IN PARALLEL IN CURRENT MODE!
This creates a short circuit → blown fuse or damaged meter

EXAMPLE: Measuring Servo Current
You want to know how much current a servo draws:
Original: UBEC (+5V) — Servo Red Wire
Modified: UBEC (+5V) — Multimeter RED (A jack) — Multimeter BLACK — Servo Red Wire
Ground connection unchanged: UBEC GND — Servo Black Wire
Turn on power, move servo through full range:

- Reading @ rest: 50 mA
- Reading @ moving: 300 mA
- Reading @ stall: 800 mA ← This is what you need for budgeting!

SAFER ALTERNATIVE: Use a "current sensor module" or "power analyzer"

- Inline device that measures current without breaking circuit
- Safer for beginners
- Can log current over time

3.4 Measuring Resistance (and Continuity)

Resistance is measured with POWER OFF

PROCEDURE:

1. Disconnect component from circuit (or at least one end)
2. Set dial to "Ω" (ohms)
3. Select range (200Ω, 2kΩ, 20kΩ, etc.)
4. Touch probes to component leads
5. Read resistance value

WHY DISCONNECT FROM CIRCUIT?
Other components in parallel will affect the reading!

EXAMPLE 1: Testing a Resistor
Resistor labeled "220Ω":

- Measured: 217Ω ← Within tolerance (usually ±5%)

Resistor labeled "1kΩ" (1000Ω):

- Measured: 1.02kΩ ← Good!

EXAMPLE 2: Testing Wire Continuity
Most useful multimeter function for UAV assembly!
Set dial to continuity mode (beeper icon)

Touch probes to both ends of a wire:

- BEEP + Low resistance ($< 1\Omega$) → Good connection
- No beep + "OL" (overload) → Broken wire!

PRACTICAL USE: After soldering connectors

- Touch RED to pin 1 on one connector
- Touch BLACK to pin 1 on other end
- Should beep → Verified connection
- Repeat for all pins

EXAMPLE 3: Checking for Short Circuits

BEFORE connecting battery:

- Set multimeter to continuity mode
- Touch RED to battery (+) connector
- Touch BLACK to battery (-) connector

Should NOT beep! (Resistance should be high)

If it beeps → SHORT CIRCUIT! Do not connect battery! Something is connecting (+) and (-) directly

3.5 Multimeter Troubleshooting Checklist

PROBLEM: "My flight controller won't turn on"

Step 1: Measure voltage AT the BEC output

- Expected: ∼5.0V
- If 0V → BEC is dead or battery is dead
- If correct → Problem is downstream

Step 2: Measure voltage AT the flight controller power pins

- Expected: ∼5.0V
- If 0V but BEC shows 5V → Bad wire or connector
- If correct → Flight controller is faulty

Step 3: Check current draw (advanced)

- If BEC voltage drops under load → Overload or short circuit

PROBLEM: "Servo twitches but won't move"

Step 1: Measure servo power supply voltage WHILE servo is moving

- Expected: ∼5.0V
- If voltage drops to 4.0V → Insufficient current, UBEC overloaded
- If stable at 5.0V → Servo may be mechanically jammed

Step 2: Measure current draw during movement

- If > UBEC rating → Need higher-current UBEC

PROBLEM: "GPS won't get satellite lock"

Step 1: Multimeter can't help much here, but check basics:

- Voltage at GPS power pins: ∼5.0V

- Continuity of TX/RX signal wires from GPS to flight controller

Usually not electrical—check:

- GPS has clear view of sky
- Antenna oriented correctly
- Wait 5+ minutes on first power-up

4 Understanding PWM Signals

4.1 What is PWM?

PWM = Pulse Width Modulation

A digital signal that controls servos, ESCs, and other devices by varying the WIDTH of pulses, not the voltage level.

SERVO/ESC PWM SIGNAL CHARACTERISTICS:

- Voltage: 5V (or 3.3V for some flight controllers)
- Frequency: 50 Hz (one pulse every 20 milliseconds)
- Pulse width: 1000 μs to 2000 μs (1 ms to 2 ms)

VISUALIZING PWM:

```
5V

    ↑                          ↑
    ← 1 ms →          (Pulse repeats every 20 ms)

    Duty cycle = 1 ms / 20 ms = 5%
```

4.2 PWM Control Values

SERVO POSITION:

Pulse Width	Servo Position
1000 μs	0° (full left)
1500 μs	90° (center)
2000 μs	180° (full right)

THROTTLE/ESC:

Pulse Width	Throttle
1000 μs	0% (motor off)
1500 μs	50% throttle
2000 μs	100% throttle (full power)

WARNING

NOTE: Some ESCs use different ranges (e.g., 1100-1900 μs). Check your ESC datasheet and calibrate!

4.3 How to Measure PWM with Multimeter

WARNING

Regular multimeters CANNOT accurately measure PWM pulse width!

However, you CAN verify signal presence:

1. Set multimeter to DC voltage
2. Touch RED to signal wire, BLACK to ground
3. Reading will show AVERAGE voltage

   Example:

   - 1 ms pulse / 20 ms period = 5% duty cycle
   - Average voltage = 5V × 0.05 = 0.25V
   - 2 ms pulse / 20 ms period = 10% duty cycle
   - Average voltage = 5V × 0.10 = 0.5V
4. If you see ∼0.25V to 0.5V → PWM signal is present
5. If you see 0V or 5V constantly → Problem!

TO ACCURATELY MEASURE PWM:

- Use an oscilloscope (expensive!)
- Use a servo tester (cheap device that displays pulse width)
- Use Arduino sketch to read pulseIn() function

4.4 PWM Troubleshooting

PROBLEM: "ESC won't arm / motor won't spin"

ESCs require a specific startup sequence:

1. Throttle stick at minimum (1000 μs)
2. Connect battery
3. Wait for ESC beeps (indicates cells detected)
4. Raise throttle slightly → Motor should spin

If motor doesn't respond:

- Check PWM signal wire is connected (usually white/yellow)
- Verify ESC is receiving power (check voltage)
- Calibrate ESC throttle range (see ESC manual)

PROBLEM: "Servo jitters or moves erratically"

CAUSES:

1. **Electrical noise on signal wire**
   - Solution: Twist signal wire with ground wire
   - Keep signal wires away from motor wires

2. **Insufficient power (voltage drop)**
   - Measure voltage at servo while it moves
   - If drops below 4.5V → Need thicker power wires or higher-current BEC
3. **Bad connection**
   - Check continuity of all three servo wires
   - Re-solder connectors

5 Grounding and Electrical Noise

5.1 Why Grounding Matters

GROUND = REFERENCE POINT for all voltages in your system

Without a proper common ground:

- Voltages are undefined
- Signals won't be read correctly
- Electrical noise causes interference
- Components may be damaged

KEY INSIGHT

GOLDEN RULE:
ALL COMPONENTS IN YOUR UAV MUST SHARE A COMMON GROUND!

5.2 Ground Loops and Noise

GROUND LOOP: Multiple paths between grounds create circulating currents

EXAMPLE OF PROBLEM:

If Ground wire A has different resistance than Ground wire B, a small voltage difference exists between the two "grounds"!

This can cause:

- ADC reading errors on Arduino
- PWM signal corruption
- Random resets or glitches

SOLUTION: Star grounding

All grounds connect to ONE central point (usually battery negative):

$$\begin{array}{c}\text{Battery (-)} \leftarrow \text{Central ground point} \\ \downarrow \\ \text{All components connect here}\end{array}$$

5.3 Sources of Electrical Noise

1. MOTOR/ESC NOISE

Brushless motors create high-frequency electrical noise

Effects:

- Corrupts GPS reception

- Causes servo jitter
- Interferes with radio receiver

SOLUTIONS:

- Keep motor wires short
- Twist motor wires together (reduces electromagnetic radiation)
- Add capacitors across motor terminals (suppresses noise)
- Separate signal wires from motor wires (at least 5 cm apart)

2. POWER SUPPLY NOISE

When servos move, they draw sudden current spikes
This causes voltage ripple on power rails

SOLUTIONS:

- Add capacitors near each component (decoupling capacitors)
  - 100 μF electrolytic capacitor on BEC output
  - 0.1 μF ceramic capacitor at each component's power pins
- Use separate BECs for servos vs. flight-critical systems (ALREADY DOING THIS!)

3. RADIO FREQUENCY INTERFERENCE (RFI)

Long wires act as antennas, picking up noise

SOLUTIONS:

- Keep wires as short as practical
- Twist signal + ground pairs together
- Use shielded cable for sensitive signals (e.g., GPS antenna)

5.4 Best Practices for Your UAV

- CONNECT all component grounds to battery negative terminal (directly or through BECs that connect to battery)
- TWIST signal wires with their ground/return wires
  - Reduces crosstalk between signals
  - Minimizes antenna effect (noise pickup)
- ROUTE motor wires AWAY from signal wires
  - If they must cross, do so at 90° angle (minimizes coupling)
- ADD a large capacitor (100-220 μF) across battery terminals
  - Smooths voltage spikes when motor draws high current

- TEST grounding by measuring voltage between grounds
  - Connect multimeter BLACK to battery (-)
  - Touch RED to ground pin of each component
  - Should read 0.00V (or $< 0.01V$)
  - If $> 0.1V \rightarrow$ Poor ground connection!

6 Systematic Troubleshooting Approach

6.1 The Troubleshooting Mindset

When something doesn't work, DON'T just try random fixes!

Follow this systematic approach:

1. **OBSERVE** - What exactly is wrong? Be specific.
2. **ISOLATE** - Which component or connection is faulty?
3. **VERIFY** - Use measurements (multimeter) to confirm your hypothesis
4. **FIX** - Make ONE change at a time
5. **TEST** - Did the fix work?

6.2 Power Troubleshooting Flowchart

SYMPTOM: "Nothing turns on"

1. **Check battery voltage (with multimeter)**
   - Is it $> 13.2V$? (3.3V/cell minimum)
   - NO → Charge battery
   - YES → Continue
2. **Check XT60 connector continuity**
   - Are positive and negative both connected?
   - NO → Resolder connector
   - YES → Continue
3. **Measure voltage at ESC input (14.8V expected)**
   - 0V → Bad Y-harness or connector
   - 14.8V → Continue
4. **Measure voltage at ESC BEC output (5V expected)**
   - 0V → ESC is dead or BEC damaged, try external UBEC alone
   - 5V → Continue
5. **Measure voltage at component power pins**
   - 0V → Bad wire from BEC to component, resolder
   - 5V → Component is faulty

6.3 Component-Specific Troubleshooting

FLIGHT CONTROLLER WON'T BOOT:

1. Verify 5V on power pins (multimeter)
2. Check current draw (should be ∼100-200 mA)
   - If 0 mA → No power reaching flight controller
   - If > 500 mA → Possible short circuit
3. Check USB connection (can it connect to computer?)
   - YES → Power issue, check external power supply
   - NO → Flight controller may be bricked, reflash firmware

GPS NO SATELLITE LOCK:

1. Verify 5V on GPS power pins
2. Check continuity of TX/RX wires to flight controller
3. Verify GPS is configured for correct baud rate (usually 38400)
4. Test outdoors with clear sky view (takes 5-15 min on first boot)
5. Check for electrical noise:
   - Move GPS farther from motor/ESC
   - Add aluminum foil shield under GPS (acts as ground plane)

SERVO NOT RESPONDING:

1. Verify 5V on servo power pins while it attempts to move
   - If drops $< 4.5V$ → Insufficient current (upgrade UBEC)
2. Test servo with servo tester (bypass flight controller)
   - Works → Problem is flight controller or wiring
   - Doesn't work → Servo is dead
3. Check PWM signal continuity from flight controller to servo
4. Verify flight controller channel is configured correctly (transmitter mixer)

MOTOR WON'T SPIN:

1. Verify 14.8V at ESC input
2. Check ESC programming (throttle calibration)
3. Test motor directly with servo tester (if 3-wire ESC)
4. Verify motor isn't mechanically jammed (try spinning by hand)
5. Check for loose motor wires (intermittent connection)

ARDUINO NOT LOGGING DATA:

1. Verify 5V power at Arduino
2. Test sensors individually (I2C scanner sketch)
3. Check SD card:
   - Is it formatted FAT32?
   - Is it inserted correctly?
   - Test write with simple Arduino sketch
4. Add Serial.print() debugging to code to see where it fails

6.4 Common Mistakes to Avoid

1. **MIXING UP VOLTAGE LEVELS**
   - Connecting 5V to 3.3V component → Damage!
   - Check datasheet for EVERY component
2. **REVERSING POLARITY**
   - Connecting battery backwards → EXPLOSION (LiPo fire!)
   - Double-check red = (+), black = (-) BEFORE connecting power
3. **FORGETTING COMMON GROUND**
   - Connecting only (+) and signal, forgetting ground wire
   - Every component needs THREE connections: +, signal, ground
4. **OVERLOADING BEC**
   - Connecting too many servos to one BEC
   - Calculate total current draw FIRST, compare to BEC rating
5. **MEASURING CURRENT IN PARALLEL**
   - Multimeter in current mode connected like a voltmeter → BOOM!
   - Current is measured IN SERIES (break the circuit)
6. **TESTING WITH PROPELLER ATTACHED**
   - Motor spins unexpectedly → Injury!
   - ALWAYS remove propeller during electrical testing

7 Practical Exercises

EXERCISE 1: Datasheet Scavenger Hunt

Find datasheets for three components in your UAV system:

1. Your ESC
2. Your Arduino/microcontroller
3. One sensor (BME280, air quality sensor, etc.)

For each datasheet, extract and record:

- Operating voltage range
- Current consumption (typical and maximum)
- Pinout diagram (draw it or save screenshot)
- I2C address (if applicable)
- One "interesting fact" from the datasheet you didn't know

DUE DATE: Before next meeting

EXERCISE 2: Multimeter Practice

Perform these measurements and record results:

1. Battery voltage:
Measured: _____ V
Calculated per-cell voltage: _____ V (divide by 4)
Status: ☐ Good ($> 3.7V$/cell) ☐ Needs charge ☐ Dead

2. USB phone charger output:
Measured: _____ V (should be ~5V)

3. Resistance of 1-meter wire:
Wire gauge: _____ AWG
Measured: _____ Ω (should be $< 1\Omega$ for short wires)

4. Continuity test on a homemade cable:
Wire 1: ☐ Beeps (good) ☐ No beep (broken)
Wire 2: ☐ Beeps (good) ☐ No beep (broken)
Wire 3: ☐ Beeps (good) ☐ No beep (broken)

DUE DATE: Complete during next lab session

EXERCISE 3: Draw Your System Schematic

Create a schematic diagram of YOUR UAV electrical system showing:

- Battery (with voltage label)

- Y-harness
- ESC and external UBEC
- Both power rails (Rail 1 and Rail 2)
- All components: Flight controller, Arduino, GPS, RC receiver, servos
- Ground connections (common ground symbol)

You can draw by hand or use:

- Draw.io (free, online)
- Fritzing (beginner-friendly)
- KiCad (professional, but complex)

HINT: Reference the schematic example in Section 1.4

DUE DATE: Before flight testing begins

EXERCISE 4: PWM Investigation

Write a simple Arduino sketch that:

1. Reads PWM input from RC receiver on pin 2
2. Prints the pulse width (in microseconds) to Serial Monitor
3. Maps the pulse width to a percentage (1000 = 0%, 2000 = 100%)

Use this to verify your RC receiver outputs correct PWM values.

TEST:
Throttle stick at minimum → Should read ∼1000 μs
Throttle stick at center → Should read ∼1500 μs
Throttle stick at maximum → Should read ∼2000 μs

Record actual measured values:
Minimum: _____ μs
Center: _____ μs
Maximum: _____ μs

ARDUINO CODE HINT:

```
int pulse = pulseIn(2, HIGH, 25000);  // Read pulse on pin 2
Serial.println(pulse);
```

DUE DATE: Whenever you receive your RC equipment

8 Advanced Topic - Decoupling Capacitors

(This section is OPTIONAL but highly recommended for understanding why some circuits include capacitors you didn't think were necessary!)

8.1 Why Decoupling Capacitors?

When a component (like a servo or Arduino) suddenly draws current, the voltage on the power rail can temporarily drop. This is called a "voltage sag" or "brownout."

CAUSES OF VOLTAGE SAG:

1. Wire resistance (even tiny amounts matter!)
2. BEC can't respond instantly to current changes
3. Inductance in long wires (resist changes in current)

SOLUTION: Capacitor acts as a "local battery"

- Stores charge near the component
- Supplies current during sudden demand
- Smooths voltage ripple

ANALOGY: Water tower near your house

- City water supply = BEC
- Pipes = Wires (have resistance)
- Water tower = Capacitor (local storage)

When you suddenly turn on all faucets, the water tower supplies water instantly while the city pumps catch up.

8.2 Types of Capacitors

ELECTROLYTIC CAPACITOR (100 μF - 1000 μF)

- Large capacitance
- Good for bulk energy storage
- Polarized (must connect (+) to (+), (-) to (-))
- Slower response time

USE: At BEC outputs, near power-hungry components

CERAMIC CAPACITOR (0.1 μF - 10 μF)

- Small capacitance

- Very fast response time
- Not polarized (can connect either way)
- Physically small

USE: Right next to IC chips (flight controller, Arduino)

8.3 Where to Add Capacitors in Your UAV

1. ACROSS BATTERY TERMINALS

- 220 μF to 470 μF electrolytic
- Reduces voltage spikes when motor draws high current
- Protects ESC and BECs

2. AT EACH BEC OUTPUT

- 100 μF electrolytic + 0.1 μF ceramic
- Smooths BEC output ripple
- Place physically close to BEC

3. AT EACH COMPONENT'S POWER PINS

- 0.1 μF ceramic
- As close as possible to IC power pins
- Flight controller and Arduino especially benefit

CONNECTION:
Always connect capacitor between (+) and (-) [or (+) and GND]

9 Preparing for Hands-On Assembly

9.1 Pre-Assembly Checklist

BEFORE you start assembling:

- ☐ All components received and inspected
- ☐ Datasheets downloaded and key specs recorded in Quick Reference Card
- ☐ Multimeter tested and working (check battery, test on known voltage)
- ☐ Soldering iron heated and tip tinned
- ☐ Wiring diagram printed and accessible
- ☐ Safety Protocol reviewed (especially LiPo handling!)
- ☐ Work area clean and organized
- ☐ Fire extinguisher or water nearby (LiPo safety)

9.2 Assembly Sequence (Recommended)

PHASE 1: Power System (DO THIS FIRST)

1. Install XT60 connectors on battery and Y-harness
2. Test continuity of Y-harness
3. Connect ESC and UBEC to Y-harness
4. Power on (no other components connected yet)
5. Verify ESC BEC outputs 5V
6. Verify UBEC outputs 5V
7. Measure current draw (should be very low, ∼10-50 mA)

WARNING

DO NOT PROCEED if voltages are incorrect!

PHASE 2: Flight-Critical Components (RAIL 1)

1. Connect flight controller to ESC BEC
2. Verify flight controller boots (LEDs light up)
3. Connect RC receiver
4. Verify RC receiver binds to transmitter
5. Connect GPS module

6. Wait for GPS satellite lock

Test: All components powered simultaneously, battery voltage $> 14V$

PHASE 3: Data Collection System

1. Connect Arduino to RAIL 1
2. Upload test sketch (blink LED or print to serial)
3. Connect sensors one at a time
4. Test each sensor individually
5. Install SD card module and test data logging

Test: Log data for 5 minutes, verify file saved correctly

PHASE 4: Servos (RAIL 2)

1. Connect ONE servo to external UBEC
2. Verify servo moves through full range
3. Measure current draw at stall
4. Connect remaining servos
5. Test all servos simultaneously
6. Verify UBEC voltage doesn't drop below 4.8V under load

Test: All servos to full deflection at once, no brownouts

PHASE 5: Motor System (SAVE FOR LAST)

1. Connect motor to ESC (NO PROP YET!)
2. Connect ESC signal wire to flight controller
3. Arm ESC (throttle low, then battery, wait for beeps)
4. Test motor at 10% throttle (should spin smoothly)
5. Measure motor current
6. If all good, install propeller (secured drone first!)

9.3 Testing Before Flight

BENCH TEST (no propeller, secured):

- ☐ All components powered on
- ☐ Flight controller boots and shows GPS lock
- ☐ RC transmitter controls all servos correctly

- ☐ Throttle stick moves (but ESC disarmed for safety)
- ☐ Data logging works continuously for 10 minutes
- ☐ Battery voltage after 10 min test: _____ V

GROUND TEST (propeller on, drone secured):

- ☐ Arm ESC
- ☐ Slowly increase throttle to 50%
- ☐ Verify motor spins in correct direction
- ☐ Check for vibrations (balance propeller if needed)
- ☐ Measure current draw: _____ A
- ☐ Verify control surfaces respond correctly
- ☐ Test failsafe (turn off transmitter → motor stops)

RANGE TEST:

- ☐ Walk 50 meters away with transmitter
- ☐ Verify all controls still respond
- ☐ Repeat at 100 meters
- ☐ If control is lost, do NOT fly! Check antenna placement.

10 Summary and Next Steps

What You Learned in This Lesson:

- How to read electrical schematics and wiring diagrams
- How to extract critical information from component datasheets
- How to use a multimeter to measure voltage, current, and continuity
- How PWM signals work and how they control servos/ESCs
- Why grounding is critical and how to avoid ground loops
- Systematic troubleshooting approach for electrical problems
- Where to add capacitors to reduce noise and voltage sags

Skills You Can Now Apply:

- Debug "component won't turn on" problems using multimeter
- Verify correct voltage at each point in your system
- Test continuity of all connections before powering on
- Identify and fix grounding issues
- Read your component datasheets to find pinouts and specs
- Systematically troubleshoot problems instead of guessing

What's Next? (Lesson 3 Preview)

The next lesson will cover:

LESSON 3: SOLDERING, CONNECTORS, AND PROFESSIONAL ASSEMBLY TECHNIQUES

Topics:

- How to solder properly (technique, temperature, inspection)
- Choosing the right connectors (XT60, JST, Dupont, bullet connectors)
- Crimping vs. soldering
- Heat shrink tubing and strain relief
- Wire management and cable routing
- Building a custom wiring harness
- Quality control and inspection
- Common soldering mistakes and how to fix them

Homework Before Next Lesson:

1. Complete Exercise 1: Datasheet Scavenger Hunt
2. Fill out Component Quick Reference Card (as components arrive)
3. Practice with multimeter (Exercise 2)
4. Review soldering safety videos online (search "soldering basics")

Questions for Review:

Test your understanding:

1. You measure 5.2V at the BEC output, but only 4.5V at the flight controller. What is the most likely cause, and how do you fix it?
2. Your servo twitches randomly. List three possible causes and how you would test for each one.
3. Why is it dangerous to measure current with the multimeter set incorrectly?
4. Your motor spins, but GPS won't get satellite lock. Could these be related? Explain.
5. What is the purpose of connecting all grounds to a common point?

(Answers will be discussed in next lesson)

CONGRATULATIONS!

You now have the foundational knowledge to read schematics, interpret datasheets, and troubleshoot electrical problems systematically.

These skills will serve you throughout this project and in future electrical engineering work.

REMEMBER: When in doubt, measure first, then act!

END OF LESSON 2